

Bo Shang, Ph.D.

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Postdoctoral Scholar at Civil Engineering, Grove School of Engineering, CUNY City College
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EXPERIENCE

- **AI & Mobility Research Lab, CUNY City College** July 2025 - Present
Postdoctoral Scholar New York, NY, USA
 - Leading research on AI-powered traffic monitoring and LiDAR-based object detection
 - Developing multimodal sensing frameworks for intelligent mobility and infrastructure safety
- **Civil Engineering, CCNY** Jan 2025 - June 2025
Graduate Research Assistant New York, NY, USA
 - Conducted research on transportation systems and AI applications
- **CCNY Robotics Lab** Dec 2022 - Dec 2024
Post-Doc Researcher New York, NY, USA
 - Developed automated bridge inspection systems integrating AI, cloud technologies, and robotics
 - Engineered CNNs for structural damage detection and deployed models on AWS cloud
- **Vaughn College** Jun 2024 - Aug 2024
Instructor, Computer Engineering Summer Academy (AI module) New York, NY, USA
 - Taught AI concepts and applications in computer engineering summer program
- **Vaughn College of Aeronautics and Technology** Jan 2024 - May 2024
Adjunct Faculty New York, NY, USA
 - Taught courses in robotics, mechanics, control, and AI principles
- **CUNY City College** Aug 2023 - Dec 2023
Adjunct Assistant Professor New York, NY, USA
 - Taught Electric Circuits (ENGR 204)
- **Missouri University of Science and Technology** Jan 2020 - Nov 2022
Post Doctoral Fellow Missouri, USA
 - Developed advanced drone systems with robotic arms for automated bridge inspections
 - Created vision-based control systems for UAV guidance and bridge inspection
- **University of California, Merced** Jan 2016 - Aug 2017
Lecturer Merced, CA, USA
 - Teaching Assistant for Mechatronics, Engineering Computing (Fortran and MATLAB), and Unmanned Aerial Systems
 - Initial Designer of laboratories for Unmanned Aerial Systems course
- **University of California, Merced** Aug 2015 - Sep 2017
Junior Specialist Merced, CA, USA
 - Conducted research on unmanned aerial systems and robotics

EDUCATION

- **Northeastern University** 2013 - 2020
Ph.D., Pattern Recognition and Intelligent Systems China
- **University of California, Merced** 2015 - 2017
Exchange Ph.D. Student Merced, CA, USA
- **Northeastern University** 2011 - 2013
M.E., Pattern Recognition and Intelligent Systems China
 - GPA: 3.53/4.00
- **Northeastern University** 2007 - 2011
B.E., Automation China
 - GPA: 3.42-3.74/4.00

PROJECTS

• **Traffic Monitoring using Fixed LiDAR**

Nov 2024 - Present

Tools: Python, Computer Vision, CNN, LiDAR Processing, Deep Learning

- Developing an end-to-end pipeline for traffic monitoring with fixed LiDAR systems, including background subtraction, object segmentation, and detection
- Working on multi-frame vehicle reconstruction to generate individual vehicle models and classify moving objects such as vehicles, motorcycles, bicycles, and pedestrians
- Evaluating various CNN-based networks for object detection from LiDAR traffic data
- Designing a flexible mechanism that enables training on one dataset and inference on another, even with different configurations

• **Advanced Bridge Inspection Automation System**

Dec 2022 - Present

Tools: Python, CNN, AWS Cloud, WebODM, iOS Development, Computer Vision, Robotics

- Spearheaded the development of a comprehensive automated system for bridge inspection, integrating cutting-edge AI, cloud technologies, and robotics
- Engineered and trained sophisticated CNNs for precise detection of structural damages, including cracks, spalling, and stains
- Successfully deployed AI models on AWS cloud, ensuring scalability and high-performance processing of inspection data
- Designed and implemented a custom WebODM-based platform that streamlines the entire inspection workflow, including automated damage segmentation, 3D reconstruction, interactive visualization, and precise measurement of cracks
- Developed a handy iOS application to control and operate a specialized climbing robot, enhancing the reach and efficiency of bridge inspections

• **Bridge Inspection Robot Deployment System (BIRDS)**

2020 - 2022

Tools: Python, C++, PID Controller, NVIDIA Jetson, Computer Vision, ROS

[\[\[Link\]\]](#)

- Developed an advanced drone system with robotic arms for automated bridge inspections
- Engineered a sophisticated PID controller to precisely manage the opening and closing mechanisms of robotic arms, ensuring optimal performance and safety during inspections
- Designed and implemented cutting-edge image processing algorithms for accurate girder detection, leveraging the computational power of NVIDIA Jetson platform
- Created a comprehensive demonstration showcasing autonomous flight, automatic clamping to bridge structures, and efficient traversal along inspection paths

• **Unmanned Aerial System of Visible Light, Infrared and Hyperspectral Cameras**

2020 - 2022

Tools: Python, Path Planning, Signal Processing, Data Analytics

[\[\[Link\]\]](#)

- Developed a multi-modal UAV system with novel signal processing and data analytics capabilities

• **Robot-assisted Underwater Acoustic Imaging for Bridge Scour Evaluation**

2020 - 2021

Tools: Python, C++, ROS, Arduino, Embedded Linux, Computer Vision, PID Controller

- Developed robotic system for underwater acoustic imaging and bridge scour evaluation

• **Drone Visual Servoing Control System**

2015 - 2019

Tools: Python, Embedded Linux, Raspberry Pi, Computer Vision, Fractional Order Controller

- Designed and implemented a drone visual servoing control system using fractional order control techniques

• **SmarCaveDrone: Sense-and-avoid and GPS-denied Navigation**

2015 - 2017

Tools: Python, Computer Vision, Navigation Systems

- Developed cave mapping UAV system with sense-and-avoid capabilities and GPS-denied navigation

- [C.1] Bo Shang, Yiqiao Li. **Roadside LiDAR for Cooperative Safety Auditing at Urban Intersections: Toward Auditable V2X Infrastructure Intelligence**. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops, DriveX Workshop on Foundation Models for Autonomous Driving (archival track), 2026.
- [S.1] Yiqiao Li, Bo Shang, Jie Wei. **Bridging the Modality Gap in Roadside LiDAR: A Training-Free Vision-Language Model Framework for Vehicle Classification**. arXiv preprint [arXiv:2602.09425](https://arxiv.org/abs/2602.09425), 2026.
- [S.2] B. Shang, Y. Li, B. Horbach, K. Villon. **Exploring Drive-By Sensing and Deep Learning for Urban Truck Characterization Using Google Street View Imagery**. SSRN preprint [5312573](https://ssrn.com/abstract=5312573), 2025.
- [C.2] Bo Shang, Yiqiao Li, Arian Golrokh Amin, Camille Kamga and Jie Wei. **AI-Enhanced Sensing for Vulnerable Road User Safety at Signalized Intersections: A Survey**. Procedia Computer Science, vol. 265 (22nd International Conference on Mobile Systems and Pervasive Computing, MobiSPC), 2025.
- [C.3] Bo Shang, Yiqiao Li, Jie Wei and Camille Kamga. **Investigating the LiDAR Beam Count Needed for Effective Vulnerable Road User Detection**. Procedia Computer Science, vol. 265 (22nd International Conference on Mobile Systems and Pervasive Computing, MobiSPC), 2025.
- [J.1] Jinglun Feng, Bo Shang, Ejup Hoxha, César Hernández, Yang He, Weihang Wang, Jizhong Xiao. **Robotic Inspection and Data Analytics to Localize and Visualize the Structural Defects of Concrete Infrastructure**. IEEE Transactions on Automation Science and Engineering, 2025 (Presented at IROS 2025).
- [J.2] Ejup Hoxha, Jinglun Feng, Agnimitra Sengupta, David Kirakosian, Yang He, Bo Shang, Ardian Gjinofci, and Jizhong Xiao. **Contrastive Learning for Robust Defect Mapping in Concrete Slabs using Impact Echo**. Construction and Building Materials (IF 7.4, cite score 13.8), 2024.
- [J.3] Zhang, Haibin, Zhenhua Shi, Liujun Li, Pu Jiao, Bo Shang, and Genda Chen. **Code-specified early delamination detection and quantification in a RC bridge deck: passive vs. active infrared thermography**. Journal of Civil Structural Health Monitoring, 2024: 1-18.
- [C.4] L. Li, B. Shang, I. Jayawardana and G. Chen, **Hardware-in-the-loop and Digital Twin Enabled Autonomous Robotics-assisted Environment Inspection**, 2023 6th International Symposium on Autonomous Systems (ISAS), Nanjing, China, 2023, pp. 1-5, doi: 10.1109/ISAS59543.2023.10164352.
- [J.4] Genda Chen*, Liujun Li, Haibin Zhang, Zhenhua Shi, Bo Shang, Derek Edwards, Daniel McDonald, Rueil Manzambi, and Joseph Ressel. **Robot-assisted, Remote Nondestructive Testing and Evaluation (rNDT&E)**. Material Design, 2023.
- [C.5] Haibin Zhang, Pu Jiao, Liujun Li, Zhenhua Shi, Bo Shang, Genda Chen, **Delamination detection of concrete bridge slab through UAV-based thermal scanning**, 8th World Conference on Structural Control and Monitoring (8WCSCM), 2022.
- [C.6] Zhenhua Shi, Bo Shang, Haibin Zhang, Liujun Li, Genda Chen. **Evaluation of User-friendliness of Several UASs in Bridge Inspection**. 8th World Conference on Structural Control and Monitoring (8WCSCM), 2022.
- [C.7] Liujun Li, Genda Chen, Bo Shang. **Mixed Reality Enabled Digital Twin for Robot-assisted Bridge element Inspection and maintenance**. 8th World Conference on Structural Control and Monitoring (8WCSCM), 2022.
- [C.8] Jiao, P., Shang, B, Li, L., and Chen, G. **The ceiling effect and flight insight of unmanned aerial vehicles during proximity inspection of bridges via computational fluid dynamics modeling and simulations**, Proceedings of the 13th International Workshop on Structural Health Monitoring, August 31 – September 2, 2021, Stanford, CA, 2021. (online on Aug 1, 2022)
- [C.9] Bo Shang, Liujun Li, Pu Jiao, Rafael Cardona Huerta, Joseph Ressel, Andrew Rawlings, Buddy Scharfenberg, and Genda Chen. **Drone vision-based clamping strategy for bridge inspection [Poster]**. INSPIRE-UTC 2021 Annual Meeting, 2021.
- [C.10] B Shang, A Reven, P Jiao, B Li, G Chen. **Vision-Based Non-GPS UAV Guidance for Bridge Inspection [Poster]**. INSPIRE-UTC 2020 Annual Meeting, 2020.
- [C.11] A Reven, P Jiao, B Shang, G Chen. **Clamping Design for Bridge Inspection Robot Deployment Systems (BIRDS) Prototype II [Poster]**. INSPIRE-UTC 2020 Annual Meeting, 2020.
- [C.12] A Reven, P Jiao, B Shang, G Chen. **Bridge Inspection Robot Deployment Systems (BIRDS) Prototype II [Slides]**. INSPIRE-UTC 2020 Annual Meeting, 2020.
- [J.5] Bo Shang, Jianxin Liu, Yunzhou Zhang, Chengdong Wu, YangQuan Chen. **Fractional Order Flight Control of Quadrotor UAS on Vision-based Precision Hovering with Larger Sampling Period**. Nonlinear Dynamics, 2019.
- [C.13] Bo Shang, Chengdong Wu, YangQuan Chen. **Neighborhood optimization method for shaping Bode plot with larger phase margin**. Proceedings of the ASME 2019 International Design Engineering Technical Conferences & Computers and Information in Engineering Conference (IDETC/CIE), 2019.

- [C.14] Bo Shang, Yunzhou Zhang, Chengdong Wu, YangQuan Chen. **Fractional Order Flight Control of Quadrotor UAS: an OS4 Benchmark Environment and a Case Study**. International Conference on Control, Automation, Robotics and Vision (ICARCV), 2018.
- [C.15] Bo Shang, Chengdong Wu, Yunzhou Zhang, YangQuan Chen. **Fractional Order Flight Control of Quadrotor UAS: A Simscape Benchmark Environment and A Case Study**. 2018 IEEE Chinese Guidance, Navigation and Control Conference (CGNCC), Xiamen, 2018, pp. 1-6.
- [C.16] Bo Shang, et al. **Analysis of Maximum Possible Sampling Period for a Real-Time Vision-Based Control System**. ASME 2017 International Design Engineering Technical Conferences and Computers and Information in Engineering Conference. American Society of Mechanical Engineers, 2017.
- [C.17] Zhang, G., Shang, B., Chen, Y., & Moyes, H. **SmartCaveDrone: 3D cave mapping using UAVs as robotic co-archaeologists**. 2017 International Conference on Unmanned Aircraft Systems (ICUAS), 2017, pp. 1052-1057.
- [C.18] B. Shang, J. Liu, T. Zhao and Y. Chen, **Fractional order robust visual servoing control of a quadrotor UAV with larger sampling period**, 2016 International Conference on Unmanned Aircraft Systems (ICUAS), Arlington, VA, 2016, pp. 1228-1234.
- [J.6] Bo Shang, Chengdong Wu, Yuchao Hu, Jianyu Yang. **An Algorithm of Visual Reconnaissance Path Planning for UAVs in Complex Spaces**. Journal of Computational Information Systems, 10(19), 2014.
- [C.19] Bo SHANG, Chengdong WU, Tingting MENG, Chengxi GAO, Yunzhou ZHANG. **A Data/Image Transmission Device Based on TCP/IP Protocol**. WiCOM2012 (International Conference on Wireless Communications, Networking and Mobile Computing), 2012.
- [C.20] Tingting Meng, Chengdong Wu, Bo Shang, Chengxi Gao, Yunzhou Zhang. **Design of point to multi-point wireless communication system based on ZigBee**. WiCOM2011 (International Conference on Wireless Communications, Networking and Mobile Computing), 2011.
- [J.7] GAO Chengxi, WU Chengdong, ZHANG Yunzhou, SHANG Bo, MENG Tingting. **Research on remote image/data transmission based on TCP/IP protocol**. Mechanical & Electrical Engineering Magazine, 2011.
- [P.1] Chengdong Wu, Bo Shang, Yunzhou Zhang, Chengxi Gao, Tingting Meng. **Data/image transmission device based on TCP/IP (Transmission Control Protocol/Internet Protocol)** (CN 102427464 B).
- [P.2] Yunzhou Zhang, Bo Shang, Chengdong Wu, Pengju Si, **Internet-based interactive digital media terminal device** (CN 102306237 A).
- [P.3] G Chen, A Reven, B Shang, Z Shi, L Li, etc. **Unmanned vehicle having flight configuration and surface traverse configuration** (U.S. Patent No. 12,296,994. 13 May 2025).

TEACHING EXPERIENCE

• ME 190: Mechatronics <i>University of California, Merced</i>	2016 - 2017
• ME 143: Unmanned Aircraft Systems <i>University of California, Merced</i>	2016 - 2017
• ME 021: Engineering Computing (Fortran and MATLAB) <i>University of California, Merced</i>	2016 - 2017
• ENGR 204: Electric Circuits <i>CUNY City College</i>	2023
• MCE 355: Robotics, Mechanics and Control <i>Vaughn College of Aeronautics and Technology</i>	2024
• SBC 012: Principles of AI <i>Vaughn College of Aeronautics and Technology</i>	2024
• SBC 014A: Principles of Research-AI <i>Vaughn College of Aeronautics and Technology</i>	2024

HONORS AND AWARDS

• Teaching Certificate <i>Association of College and University Educators (ACUE)</i> ◦ Certification in effective teaching practices for higher education	2021
• Remote Pilot Certificate for Small Unmanned Aircraft Systems <i>Federal Aviation Administration (FAA)</i> ◦ Licensed to operate small unmanned aircraft systems commercially	2016
• Financial Support for Exchange Program <i>Chinese Scholarship Council (CSC)</i> ◦ \$38.4k financial support for two-year exchange program at University of California, Merced	2015 - 2017

- **Best System Control Award [Team Leader]** 2014
International Aerial Robotics Competition, AUVSI Foundation
 - Recognized for outstanding system control implementation in international competition
- **Best Mission Planning Award [Team Leader]** 2014
International Aerial Robotics Competition, AUVSI Foundation
 - Recognized for exceptional mission planning and execution
- **Meritorious Prize [Programmer]** 2010
International Mathematical Contest in Modeling, USA
 - Awarded for outstanding performance in mathematical modeling competition
- **First Prize, Northeastern Region** 2010
National Smart Car Competition, Freescale, China
 - Regional champion in smart car design and programming competition

VOLUNTEER EXPERIENCE

- **Coach** 2022
FIRST Robotics Competition (K-12 level)
 - Mentored K-12 students in robotics design, programming, and competition strategy
- **Judge** 2024
VEX Robotics Competition (Middle, high school and college level)
 - Evaluated robot performance and design in regional robotics competition
- **Tutor** 2023 - 2024
High School Research Assistant Program at CCNY
 - Guided high school students in research methodology and scientific writing
- **Session Chair** 2016
International Conference on Unmanned Aircraft Systems (ICUAS) [[Link]]
 - Organized and moderated conference sessions on unmanned aerial systems

REVIEWER CONTRIBUTIONS

Web of Science Profile

Nonlinear Dynamics <http://www.springer.com/engineering/mechanics/journal/11071>
International Conference on Unmanned Aircraft Systems <http://www.uasconferences.com/>
Journal of Intelligent & Robotic Systems <http://www.editorialmanager.com/jint/default.aspx>
ISA Transactions <https://ees.elsevier.com/isatrans/mainpage.html>
IEEE Transactions on Control Systems Technology <http://www.ieeecss.org/publications/tcst>
Intelligent Buildings International <https://mc.manuscriptcentral.com/inbi>
IEEE International Conference on Multisensor Fusion and Integration for Intelligent Systems
International Journal of Advanced Robotic Systems
<https://us.sagepub.com/en-us/nam/international-journal-of-advanced-robotic-systems/journal202567>
International Conference on Robotics and Automation
<http://www.ieee-ras.org/conferences-workshops/fully-sponsored/icra>
Control Engineering Practice <https://www.journals.elsevier.com/control-engineering-practice>
IET Control Theory and Applications <http://digital-library.theiet.org/content/journals/iet-cta>
Mechatronics <https://www.journals.elsevier.com/mechatronics>

CERTIFICATIONS

- **Remote Pilot Certificate for Small Unmanned Aircraft Systems**, Federal Aviation Administration (FAA) 2016
- **Teaching Certificate**, Association of College and University Educators (ACUE) 2021

ADDITIONAL INFORMATION

Research Focus: AI-powered traffic monitoring, LiDAR-vision fusion for object detection, multimodal sensing and deep learning frameworks, autonomous robotic decision-making, drone-based infrastructure inspection, bridge condition assessment, vulnerable road user monitoring, data analytics for structural health monitoring

REFERENCES

Available upon request.